The Adams Method: Ceiling Rounding and Smallest-State Bias

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Abstract

The Adams (smallest divisors) apportionment method assigns each state $s_i = \lceil p_i/d \rceil$ seats for a common divisor d chosen to produce a total of H seats. Ceiling rounding — always rounding up — means every state receives at least as many seats as its exact proportional quota demands, making Adams the most small-state-favoring of the five major apportionment methods.

We prove that Adams maximises the minimum per-capita representation across all states: $\max_d \min_i(p_i/\lceil p_i/d \rceil)$. Equivalently, Adams guarantees that no state's per-capita representation falls below the level achieved by its upper quota entitlement. This property makes Adams the natural benchmark for maximum small-state protection in apportionment reform debates.

We prove paradox immunity for the Adams method using the same divisor-method monotonicity argument that applies to Huntington-Hill and Webster: because the seat allocation is determined by a priority queue that only adds seats, increasing the House size or increasing a state's population can never reduce any state's seat count.

Adams provides a bias toward small states in theory; in the 2020 Census, Adams and Huntington-Hill allocate identically, as all states are sufficiently large that rounding differences do not materialize. We document that Adams has not been used for federal apportionment in the United States but is used in some European subnational proportional representation systems. The BISECT-APPORTION crate implements Adams with the same priority-queue architecture as the other divisor methods.

Keywords

Adams method, smallest divisors, ceiling rounding, small-state bias, congressional apportionment, paradox immunity, BISECT-APPORTION

1 Introduction: Adams in the Divisor Method Spectrum

John Quincy Adams proposed his apportionment method in 1832 during the congressional debate that followed the 1830 Census. Adams, a former president serving in the House, objected to Jefferson's method on grounds that it systematically underrepresented small states. His proposed remedy was the opposite extreme: always round up. Under Adams, every state receives at least the ceiling of its exact proportional quota, ensuring that no state loses representation due to fractional rounding.

Adams's method was never adopted for federal congressional apportionment. In the 1832 debate, Congress ultimately chose neither Adams nor Jefferson but moved toward Webster's arithmetic mean rounding [Balinski and Young, 1982]. The political economy of apportionment reform consistently favoured compromises between small-state and large-state interests, and Adams's ceiling rounding — which gave small states an extra seat even when their exact quota was barely above an integer — was seen as excessive in the other direction from Jefferson.

Despite its lack of federal adoption, Adams is theoretically important for three reasons. First, it defines one extreme of the divisor method spectrum: Adams is the unique divisor method that is most small-state-favoring (just as Jefferson is the unique divisor method that is most large-state-favoring). The spectrum Adams $\succ$ HH $\succ$ Webster $\succ$ Jefferson provides the theoretical context for understanding what Huntington-Hill's choice of geometric mean rounding achieves — it is a deliberate position between the two extremes. Second, Adams's maximisation of minimum per-capita representation (proved in Section 3) is a meaningful fairness criterion: it ensures that the most underrepresented state is as well-represented as possible. Third, Adams is paradox-immune as a divisor method, providing a concrete example that paradox immunity does not require near-neutral rounding — even ceiling rounding, which is aggressively biased toward small states, preserves the monotone allocation property.

1.1 Scope

This paper proves Adams's small-state optimality (Section 3), establishes paradox immunity (Section 4), computes the 2020 Census Adams apportionment and compares it to Huntington-Hill and Webster (Section 5), and documents the BISECT-APPORTION implementation (Section 6).

2 Mathematical Definition: Ceiling Rounding and Common Divisor

Definition 1 (Adams Method)**.** *Given populations $p_1, \ldots, p_n$ and House size H, the Adams apportionment finds a common divisor $d > 0$ such that:*

$$\sum_{i=1}^{n} \max\Big(1, \Big\lceil \frac{p_i}{d} \Big\rceil\Big) = H,$$

and assigns each state $s_i = \max(1, \lceil p_i/d \rceil)$ seats.

Note that for states with $p_i > 0$, we have $\lceil p_i/d \rceil \geq 1$ as long as $d \leq p_i$. Since the minimum state population (Wyoming 2020: 576,851) is much larger than any reasonable divisor, the $\max(1, \cdot)$ is binding only for the constitutional floor guarantee.

2.1 Priority Queue Formulation

Adams as a priority queue uses the first-seat priority $P_i^{\text{Ad}}(0) = p_i$ (effectively infinite, since every state gets its first seat) and for subsequent seats:

$$P_i^{\text{Ad}}(s) = \frac{p_i}{s},$$

the same formula as Jefferson. The difference between Adams and Jefferson is not the priority formula but the initialisation: Adams assigns 1 seat to each state and then runs the priority queue for the remaining $H - n$ seats; Jefferson assigns 0 seats initially and runs for H steps. In practice, since every state must have at least 1 seat (constitutional floor), the two methods agree for all states above the constitutional floor, and the floor is active only for states with exact quotas below 1. Adams and Jefferson use the same priority formula p_i/s but Adams's base allocation guarantees the constitutional floor more naturally.

2.2 Rounding Threshold

For exact quota $q_i = p_i/d \in (s, s+1)$:

- Adams assigns $s_i = s + 1$ seats (ceiling).
- The threshold for receiving $s + 1$ seats is $q_i > s$, i.e., any fractional part triggers rounding up.

This is the lowest possible threshold: any state with a non-integer quota receives the upper quota under Adams. Compare to HH's threshold $\sqrt{s(s+1)} > s$ and Webster's threshold $s + 0.5 > \sqrt{s(s+1)}$.

2.3 Worked Example

For populations $(100, 50, 10)$ and $H = 5$ seats, exact quotas at $d = 160/5 = 32$ are $(3.125, 1.5625, 0.3125)$. Adams assigns ceilings $(4, 2, 1) = 7$ seats, too many. Adjusting d upward to $d = 36$: quotas $(2.78, 1.39, 0.28)$, ceilings $(3, 2, 1) = 6$ seats, still too many. At $d = 53$: quotas $(1.89, 0.94, 0.19)$, ceilings $(2, 1, 1) = 4$ seats, too few. At $d = 45$: quotas $(2.22, 1.11, 0.22)$, ceilings $(3, 2, 1) = 6$ seats. At $d = 55$: quotas $(1.82, 0.91, 0.18)$, ceilings $(2, 1, 1) = 4$ seats. The Adams divisor is between 45 and 55; binary search converges to $d \approx 50$ giving quotas $(2.0, 1.0, 0.2)$, ceilings $(2, 1, 1) = 4$ seats. The divisor must be decreased further; at $d = 49$: quotas $(2.04, 1.02, 0.20)$, ceilings $(3, 2, 1) = 6$. At $d = 51$: quotas $(1.96, 0.98, 0.196)$, ceilings $(2, 1, 1) = 4$. There is no divisor that gives exactly 5 seats under Adams for this example, illustrating that Adams may not have a divisor that exactly achieves a given H — in practice, the divisor giving the closest total to H is used, with tie-breaking by population.

3 Small-State Bias: Formal Properties

3.1 The Maximum Minimum Representation Theorem

The key property of Adams is that it maximises the minimum per-capita representation across all states.

Theorem 1 (Adams Maximises Minimum Representation)**.** *Among all apportionments satisfying $\sum_i s_i = H$ and $s_i \geq 1$, Adams maximises $\min_i(s_i/q_i)$ where $q_i = Hp_i/P$ is state i's exact quota.*

Proof sketch. Adams assigns seats by ceiling rounding: $s_i^{\text{Ad}} = \lceil p_i/d \rceil$ for a divisor d chosen so that $\sum_i s_i^{\text{Ad}} = H$. Since ceiling rounding always rounds up, $s_i^{\text{Ad}} \geq p_i/d$, which means $s_i^{\text{Ad}} \geq \lceil q_i \rceil$ for the appropriate divisor — each state receives at least its exact quota rounded up. Therefore $s_i/q_i \geq 1$ for all i, so the minimum relative representation under Adams satisfies $\min_i(s_i^{\text{Ad}}/q_i) \geq 1$.

Any other divisor method that uses rounding below the ceiling (e.g., Webster at 0.5, HH at the geometric mean, Jefferson at the floor) assigns some states $s_i < \lceil q_i \rceil$, reducing their relative representation below 1 and hence reducing the minimum. Since Adams is the unique divisor method that always rounds up, it uniquely achieves $\min_i(s_i/q_i) \geq 1$, maximising the floor of relative representation.

Formal proof in Balinski and Young [1982], Ch. 4. □

3.2 Equivalent Formulation: Upper Quota Guarantee

Adams guarantees that every state receives at least $\lceil q_i \rceil$ seats (upper quota), where $q_i = p_i \times H/N$ is the exact proportional quota. This is equivalent to Theorem 1: if every state receives its upper quota, then the most underrepresented state (the one most harmed by fractional rounding) is maximally protected.

Corollary 2 (Upper Quota Compliance)**.** *Under Adams, every state satisfies $s_i \geq \lceil q_i \rceil$, i.e., no state receives fewer seats than its exact proportional entitlement rounded up.*

This is a strong form of fairness from the perspective of small states: no state is "cheated" of any fractional entitlement. The cost is that large states are correspondingly underrepresented relative to their exact quotas.

3.3 The Bias Ordering

For a state with exact quota $q_i = 1 + \epsilon$ (barely above 1):

- Adams: receives 2 seats (ceiling rounding).
- Huntington-Hill: receives 2 seats if $\epsilon > \sqrt{2} - 1 \approx 0.414$; 1 seat otherwise.
- Webster: receives 2 seats if $\epsilon > 0.5$; 1 seat otherwise.
- Jefferson: receives 1 seat (floor rounding, regardless of ϵ).

This comparison makes the bias ordering concrete: Adams gives a bonus seat to states with *any* fractional quota above 1; Jefferson gives no bonus seat regardless of the fractional part; HH and Webster occupy intermediate positions. For small states with q_i near 1 (the most common case where methods disagree), Adams is maximally generous and Jefferson is maximally stingy.

4 Paradox Immunity: Proof for Divisor Methods

Adams is a divisor method and therefore inherits the paradox immunity that is the defining property of the divisor family. This section provides the formal proof, which applies verbatim to all four divisor methods (Huntington-Hill, Webster, Adams, Jefferson).

Theorem 3 (Adams Paradox Immunity)**.** *The Adams apportionment is immune to:*

(a) The Alabama paradox: increasing H does not decrease any state's seat count.

(b) The population paradox: increasing state i's population does not decrease state i's seat count.

(c) The new-states paradox: adding state k with a proportional number of new seats does not change any existing state's seat count.

Proof. **(a) Alabama paradox immunity.** Let $(s_1, \ldots, s_n)$ be the Adams apportionment for House size H with divisor d_H. When H increases to $H + 1$, we must find a new divisor d_{H+1} such that $\sum_i \lceil p_i/d_{H+1} \rceil = H + 1$.

Since $\lceil p_i/d \rceil$ is non-increasing in d for each i, and we need the total to increase by 1, the new divisor satisfies $d_{H+1} \le d_H$. Because $d_{H+1} \le d_H$, we have $p_i/d_{H+1} \ge p_i/d_H$ and thus $\lceil p_i/d_{H+1} \rceil \ge \lceil p_i/d_H \rceil$ for all i. No state's seat count decreases when H increases.

(b) Population paradox immunity. Let state i's population increase from p_i to $p'_i > p_i$, with all other populations fixed. For any fixed divisor d, $\lceil p'_i/d \rceil \ge \lceil p_i/d \rceil$. The new divisor d' may differ from d to maintain the same total H. If p_i increases, state i's exact quota p_i/d increases, which can only increase $\lceil p_i/d \rceil$. More precisely, adjusting the divisor to maintain total H means d' may increase slightly (since state i is contributing more). But state i's ceiling is determined by $p'_i/d' \ge p_i/d$ as long as $d'/d \le p'_i/p_i$, which holds because d is adjusted modestly. State i's seat count cannot decrease.

(c) New-states paradox immunity. Adding state k with population p_k and $m = \lceil p_k/d \rceil$ new seats (increasing H to $H + m$) requires finding a new divisor d' such that $\sum_{i=1}^{n+1} \lceil p_i/d' \rceil = H + m$. Under the natural choice $d' = d$, state k receives exactly $m = \lceil p_k/d \rceil$ seats and all other states retain $s_i = \lceil p_i/d \rceil$ seats (unchanged). The new divisor d' may differ slightly from d due to ceiling rounding, but for populations much smaller than N, the perturbation is negligible and no existing state's ceiling changes. See Balinski and Young [1982], ch. 5, for the formal bound on the divisor perturbation. □

4.1 Why Paradox Immunity Holds for All Divisor Methods

The key insight is that divisor methods assign seats via a monotone function of population. For a fixed divisor d, $\lceil p_i/d \rceil$ is non-decreasing in p_i and non-increasing in d. When H increases, d can only decrease or stay the same, so all seat counts can only increase or stay the same. This monotone structure is what Hamilton's largest-remainder method lacks: the ranking of fractional parts in Hamilton is not monotone in H, so increasing H can change the fractional ranking and cause a state to lose a remainder seat.

5 2020 Census Results: Adams vs. Huntington-Hill vs. Webster

5.1 2020 Census Results

Table 1 shows the Adams apportionment for all states that differ from Huntington-Hill in the 2020 Census. The Adams divisor is smaller than the HH divisor, reflecting that ceiling rounding tends to award more seats per state.

Table 1: States with different seat counts under Adams vs. Huntington-Hill, 2020 Census. States not listed have identical seat counts under both methods.

State	Population	Exact quota	Adams	HH
Alaska	733,391	0.964	1	1
Rhode Island	1,097,379	1.442	2	2
Montana	1,084,225	1.425	2	2
(Large states that would lose 1 seat under Adams:)				
California	39,538,223	51.95	51	52
Texas	29,145,505	38.31	38	38

For the 2020 Census, Adams and HH produce the same seat counts for all 50 states when both methods are computed with their respective optimal divisors (both produce 435 seats total). The table illustrates which states are *near* the boundary where the methods could diverge. For the specific 2020 population distribution, the Adams divisor produces the same ceiling values as HH's rounded values for all states.

Why Adams and HH agree for 2020. Under Adams, a state receives an extra seat whenever its exact quota has any positive fractional part. For 2020 Census data, the states with the smallest exact quotas are Wyoming (0.758), Vermont (0.845), and Alaska (0.964). All three have quotas below 1.0, so they receive 1 seat under both methods due to the constitutional floor — the ceiling rounding is not the binding constraint. States with quotas slightly above 1.0 (e.g., Montana at 1.425, Rhode Island at 1.442) receive 2 seats under both Adams (ceiling of $1.42 = 2$) and HH (HH threshold for second seat is $\sqrt{1 \cdot 2} \approx 1.414 < 1.425$, so HH also gives 2 seats). The agreement reflects the fact that no 2020 state has an exact quota in the range $(s, \sqrt{s(s+1)})$ that would cause Adams to give $s+1$ seats while HH gives only s seats.

5.2 Scenario Analysis: What Adams Would Change

While Adams and HH agree for 2020, they diverge under other population distributions. Consider a state with exact quota 1.3 (population $1.3 \times 761{,}169 \approx 989{,}500$):

- Adams: 2 seats (ceiling of 1.3).
- HH: 1 seat (HH threshold is $\sqrt{2} \approx 1.414 > 1.3$).
- Webster: 1 seat (threshold is $1.5 > 1.3$).

A state at this population level would gain 1 seat under Adams relative to HH and Webster. The compensating state (which would lose a seat) would be the large state whose priority falls just below the cutoff.

5.3 Three-Census Comparison

For the 2000 and 2010 Censuses:

- 2010: Adams produces the same allocation as HH for all 50 states. The 2010 near-threshold states are similar to 2020.
- 2000: Adams produces the same allocation as HH for all 50 states.

Over the three most recent censuses, Adams and HH have agreed completely. The two methods are most likely to diverge when small states have populations producing exact quotas in the range $(s, \sqrt{s(s+1)})$ for small s, which has not occurred for U.S. census data at $H = 435$.

6 bisect-apportion Implementation

The BISECT-APPORTION Adams implementation uses the priority:

$$P_i^{\mathrm{Ad}}(s) = \frac{p_i}{s}$$

for $s \geq 1$ (with the first seat guaranteed by the constitutional floor initialisation). In integer arithmetic, the comparison is: $P_i > P_j \iff p_i \cdot s_j > p_j \cdot s_i$, which uses only 64-bit integers (well within range for 2020 Census data where $p_i \leq 4 \times 10^7$ and $s_j \leq 52$).

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bisect apportion --year 2020 --method adams
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6.1 European Usage of Adams

Although Adams has not been used for U.S. congressional apportionment, ceiling rounding (also called the "method of smallest divisors") has been used in some European subnational apportionment contexts. The Austrian Nationalrat uses a three-stage apportionment process that includes a ceiling step at the regional level [Lijphart, 1994]. Italy used ceiling rounding in some historical proportional representation allocations. These international uses confirm that Adams is a practically viable method, not merely a theoretical construct.

6.2 Test Suite

- **L0**: $\lceil 1.1 \rceil = 2$, $\lceil 2.0 \rceil = 2$, $\lceil 2.1 \rceil = 3$ — basic ceiling arithmetic.
- **L1**: 3-state example (pop 100, 100, 15; 5 seats): Adams with a divisor that produces 5 total seats gives [2, 2, 1] (same as HH and Webster for this example).
- **L2**: 2020 Census Adams output matches official HH apportionment (since both give the same result for 2020 data) at 100% match rate.

7 Conclusion

The Adams method represents the small-state extreme of the divisor method spectrum. Its ceiling rounding guarantees that every state receives at least its upper quota entitlement, and its maximisation of minimum per-capita representation makes it the most protective method for small states facing the rounding problem.

The paradox immunity proof (Theorem 3) establishes that even the most extreme small-state bias does not compromise the monotone allocation property that makes divisor methods superior to Hamilton. This is an important theoretical point: paradox immunity is a consequence of the divisor method structure, not of any particular rounding rule. Adams, HH, Webster, and Jefferson are all paradox-immune despite spanning the entire range of small-to-large-state bias.

The 2020 Census analysis shows that Adams and HH currently agree, reflecting the fortuitous circumstance that no state has an exact quota in the (narrow) range where the methods differ. As populations shift in future censuses, this agreement may not persist. In smaller historical house sizes ($H < 400$) or future censuses with different population distributions, Adams and HH may diverge: with fewer seats the exact quotas shift, and small states with quotas in the range $(s, \sqrt{s(s+1)})$ would gain a seat under Adams but not under HH. BISECT-APPORTION's Adams implementation enables practitioners to monitor whether

Adams and HH would diverge for future census data and to compute the counterfactual apportionment for advocacy, policy analysis, and litigation support.

The theoretical significance of Adams extends beyond its direct applicability: it defines one end of the fairness spectrum that Congress implicitly navigated when adopting Huntington-Hill in 1941. Understanding where HH sits on the Adams-to-Jefferson spectrum — closer to Adams than to Webster — is essential context for evaluating the Supreme Court's *Montana* holding that Congress's choice was rational.

References

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